

LAITH ASSAF

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EDUCATION

Michigan State University, College of Engineering

East Lansing, MI

BS Computer Science, AI/ML | Dean's List x3

August 2022 - May 2027

- **Relevant Coursework:** Data Structures & Algorithms (CSE 331), Software Design (CSE 335), Computer Systems (CSE 325), Computer Organization & ARM Assembly (CSE 320)

EXPERIENCE

PC Technician Intern

May 2026 - Present

Onsite Support Team, Delta Dental

Okemos, MI

- Deployed and configured **150+** workstations (monitors, docking stations, keyboards, mice) across a new corporate office buildout, imaging endpoints via **Ivanti** MDM to enforce standardized enterprise configuration
- Developing an automated ticket creation workflow in **Power Automate** to streamline IT ops; performed PC imaging and disk backups and drive kills across endpoint lifecycle tasks

IT and Media Services Support Associate

May 2023 - Present

MSU College of Osteopathic Medicine

East Lansing, MI

- Supported 100+ classroom AV and conferencing systems across 4 buildings serving **1,200+ students**, resolving 50-70 tickets weekly and maintaining **99.5% uptime** through a documented diagnostic methodology

PROJECTS

Bonfire Platform & Ember Framework | *TypeScript, Socket.io, Next.js*

December 2025 – April 2026

- Designed and published **Ember**, an open-source TypeScript framework for multiplayer party games (@**bonfire-ember/core/server/client** npm packages), implementing phase state machine, WebSocket room orchestration, and player lifecycle into a 5-hook API enabling **80%+ code reuse** across new game builds
- Architected a **multi-room orchestration system** on Express + **Socket.io** namespaces with **Firestore Realtime DB** state sync, hosting 4 concurrent games on a single **Railway** server with configurable disconnect strategies and TTL-based room cleanup

Pokemon Battle Agent - Reinforcement Learning | *PyTorch, PPO, MCTS*

May 2026 – Present

- Trained a competitive battle agent through **behavior cloning** on human replays, **anchored PPO** fine-tuning of an MLP policy/value network over structured per-Pokemon token observations, and **determinized MCTS** at decision time, reaching a **93% win rate** against scripted baselines after testing and retiring a transformer policy
- Built a parallel simulation and evaluation harness on the **Pokemon Showdown** engine spanning **747 rated games**; a self-run code review uncovered four confounds inflating prior results, and I restated every affected number rather than reporting the inflated figure

SoundSense - Real-Time Audio Classification | *YAMNet, Flask* | **2nd Place, MHacks**

October 2025

- Placed **2nd of 300+ participants at MHacks 2025** building a real-time audio classifier on **YAMNet** CNN and MemryX edge accelerators, hitting **95% accuracy** at <50ms latency and streaming hazard alerts through a **Flask** + **WebSocket** pipeline to a live interface for hearing-impaired users, running all inference on-device

LEADERSHIP

Michigan State University | AI Club

East Lansing, MI

Workshop Coordinator & Project Lead

January 2025 - Present

- Delivered 8+ technical workshops on **local LLM deployment** (Ollama, LM Studio), agent orchestration, and **PyTorch** fundamentals to 500+ attendees, and led a 4-student team shipping an **LLM-powered** academic performance predictor across 50+ courses on the GPT-4 API with agile sprints

TECHNICAL SKILLS

Languages: Python, C++, C#, TypeScript, SQL

AI/ML: PyTorch, reinforcement learning (PPO), MCTS, LLM APIs (Anthropic, OpenAI), Ollama

Web & Backend: FastAPI, Flask, Express, Node.js, React, Next.js, Socket.io, REST APIs, WebSockets, Firebase

Infrastructure & Tools: Git, Linux, Docker, PostgreSQL, Railway, Vercel, Claude Code, agent orchestration, MCP